The transversal cancellation identity (Q576): a degree–divisibility squeeze, and why its converse is false

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Provenance (PROTOCOL §2.3).
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Repository: `clio-vega/proofs`. **Commit:** `a7a787a9fa` (full: `a7a787a9fa61627609ed5c6c5719b24921230de8`) — this `.tex` and the verification code (`proofs/code-1024c2-q565/`) are contained in that commit, which was made *before* this block was filled in; the hash was then read off the repository with `git rev-parse` and pasted here, and checked to resolve on the remote. Nothing in this field is deferred to a later action.
Registry: `proofs/registry/rick-q488-converse.json`, subtree `clio-Q576-transversal-cancellation`.
Supersedes: `proofs/2026-10-24-Q565-zero-locus-in-M.tex`, Conjecture 11 (`conj:transversal`) — *refuted* here, in one direction, and *proved and generalised* in the other.

Abstract

This morning's Q565 artifact closed with one open conjecture (`conj:transversal`): for $\theta = (0, 1, \dots, r-1)$ and M reciprocal-free consisting of λ negation pairs $\{v_i, -v_i\}$ together with f unpaired elements, $\Psi^C_{(1^r),\theta}(M) = 0$ *if and only if* $r \geq f + 2$; it was measured 26/26 cells in both directions and the sign bookkeeping was the missing step. Both halves turn out to be wrong in instructive ways, and the repaired statement is stronger than the conjecture.

The "if" direction is now **proved** (Theorem 14), for a scope much wider than $\theta = (0, \dots, r-1)$ and with a threshold one step lower than conjectured on half of that scope: whenever the ρ-shifts $m_j = \theta_j + j - 1$ are *all of the same parity* δ, the vanishing holds for $r \geq f + 2 - \delta$. The proof is not a sign-reversing involution: both natural candidates fail, the single-pair flip because it has fixed points as soon as $r \leq \lambda + f - 1$, and the global flip because its sign is $+1$ whenever $\binom{r}{2}$ is even. Instead the $2^{|K|}$ terms of each orbit of the pair-flipping group $(\mathbb{Z}/2)^K$ are shown to sum to a rational function that is *simultaneously* divisible by $\prod_{\alpha<\beta}(x_\alpha^2 - x_\beta^2)$ and of too low a degree to be (Proposition 10). The threshold is exactly where that squeeze closes; and when all the m_j are odd the orbit sum is in addition odd in each variable, which supplies one more factor and lowers the threshold by one. The $\delta = 1$ half was invisible to the original sweep, which only ever used $\theta = (0, \dots, r-1)$, i.e. $m_j = 2(j-1)$: all even.

The "only if" direction is **false**, and the reason is a category error rather than a missing case: $r \geq f + 2$ is the threshold for vanishing *identically on the stratum*, and the conjecture read it as a pointwise criterion. Four exact rational witnesses (Theorem 18) have $r = 1 < f + 2$ and $\Psi^C = 0$ with every term nonzero; by Theorem C of the morning artifact their locus is $\prod_t w_t = \pm 1$, a codimension-one subvariety *of the stratum*, which a sampler drawing random points of the stratum can never meet. One of the four lies in a cell the 26/26 sweep actually visited.

1 What was conjectured, and the shape of the error

Conjecture 1 (`conj:transversal`, 2026-10-24 cycle 1, now settled)**.** Let $\theta = (0, 1, \dots, r-1)$ and let M be admissible and reciprocal-free, consisting of λ disjoint negation pairs $\{v_i, -v_i\}$ together with f unpaired elements. Then $\Psi^C_{(1^r),\theta}(M) = 0$ if and only if $r \geq f + 2$.

Two separate things were wrong, and they are worth naming before the mathematics because they are the same two faults the morning session diagnosed in its own predecessor, one level up.

(i) The sweep could not see the counterexamples, because it drew points. The 26/26 was measured by drawing random (v, u) from the stratum and evaluating. A statement of the form "Ψ^C vanishes at this point" has two very different causes: Ψ^C may vanish identically on the stratum, or the point may lie on a proper subvariety of the stratum. Random draws see the first and miss the second with probability 1. So the sweep was measuring the identical-vanishing threshold — which it got right — and reporting it as a pointwise criterion, which it is not. The morning artifact's own §1 says *a sampler's support is a hypothesis whether or not it is written as one*; the present fault is the next one along: *a sampler's **measure** is a hypothesis too, and it makes every positive-codimension locus invisible.* A sweep over a stratum can only ever confirm identities.

(ii) "The missing step is the sign" named the wrong missing step. The conjecture was recorded with the remark that the involution reversing one negation pair pairs the surviving terms and only the sign bookkeeping was missing. That diagnosis was wrong twice over. The single-pair flip σ_i is *not* fixed-point free: the orbit of a transversal T that does not meet the pair $\{v_i, -v_i\}$ is a fixed point, and such T exist as soon as $r \leq \lambda + f - 1$. And the global flip $\varepsilon \mapsto -\varepsilon$, which *is* fixed-point free, multiplies an orbit term by $(-1)^{\binom{L}{2}}$ when $f = 0$ — so it proves nothing when $\binom{L}{2}$ is even, i.e. for $L \equiv 0, 1 \pmod 4$, and the cell $\lambda = f + 4 = 4,\ f = 0,\ r = 4$ is measured zero with 16 nonzero terms. There is no sign-reversing involution to find. The mechanism is a degree count.

2 Setting

Convention 2. $M = \{w_1, \dots, w_n\}$ is a set of n distinct nonzero elements of a field k of characteristic 0 (*admissible*). M is *reciprocal-free* if $w_a w_b \neq 1$ for all $a \leq b$ — in particular $w_a \neq \pm 1$. For $\theta \in \mathbb{Z}^r$ put $m_j = \theta_j + j - 1$, assumed pairwise distinct. I use throughout the alternant expansion of 2026-10-22 and the pair form of $B(T)$ of this morning:

$$\Psi^C_{(1^r),\theta}(M) = \sum_{T \subseteq M,\ |T|=r} B(T)\, A(T), \qquad A(T) = \frac{\det\big(x_i^{m_j}\big)_{i,j=1}^{r}}{\prod_{i<k}(x_k - x_i)}, \tag{1}$$

$$B(T) = \frac{N(T)}{D(T)}, \qquad N(T) = \prod_{x \in T}(x^2 - 1) \prod_{\{x,y\} \subseteq T} (xy - 1) \prod_{x \in T} \prod_{t \in M \setminus T} (xt - 1), \tag{2}$$

with $T = \{x_1, \dots, x_r\}$ and $D(T) = e^r \prod_{x \in T} x \prod_{x \in T} \prod_{t \in M \setminus T} (x - t)$, $e = \prod_{t \in M} t$.

Convention 3 (the negation stratum)**.** Given $\lambda, f \geq 0$ let $\Sigma_{\lambda,f} \subseteq \mathbb{A}^{\lambda+f}$ be the set of $(v_1, \dots, v_\lambda, u_1, \dots, u_f)$ for which

$$M(v, u) \;=\; \{v_1, -v_1, \dots, v_\lambda, -v_\lambda\} \;\cup\; \{u_1, \dots, u_f\}$$

is admissible and has *exactly* λ negation pairs, i.e. $u_p \neq -u_q$ for all p, q and $u_p \neq \pm v_i$ for all p, i. Thus $n = 2\lambda + f$. Write $\Sigma^\circ_{\lambda,f}$ for the reciprocal-free part. Both are nonempty Zariski-open subsets of $\mathbb{A}^{\lambda+f}$, hence irreducible and dense in it.

3 The parity hypothesis, and what it buys

Lemma 4 (alternant parity split)**.** *Suppose $m_j \equiv \delta \pmod 2$ for all j, $\delta \in \{0,1\}$, and write $m_j = 2k_j + \delta$. Then for $T = \{x_1, \dots, x_r\}$,*

$$A(T) \;=\; \Big(\prod_{i=1}^{r} x_i\Big)^{\delta} \cdot s_\mu\big(x_1^2, \dots, x_r^2\big) \cdot \prod_{1 \le i < i' \le r} (x_i + x_{i'}),$$

where $s_\mu(y) = \det(y_i^{k_j}) / \prod_{i<i'}(y_{i'} - y_i)$ is the Schur function of the partition μ with $\mu_{r+1-j} = k_j - (j-1)$.

Proof. $\det(x_i^{2k_j+\delta}) = \big(\prod_i x_i^{\delta}\big) \det\big((x_i^2)^{k_j}\big)$, by pulling x_i^{δ} out of row i. Also $\prod_{i<i'}(x_{i'}^2 - x_i^2) = \prod_{i<i'}(x_{i'} - x_i) \prod_{i<i'}(x_{i'} + x_i)$. Divide the first display by $\prod_{i<i'}(x_{i'} - x_i)$ and insert the second. □

Verified 216/216 exactly, $r \le 4$, random rational x with distinct $|x_i|$ (`t7_parity.py`).

Corollary 5. *Under the same parity hypothesis, $A(T) = 0$ whenever T contains both members of a negation pair.*

Proof. If $x, -x \in T$ then the corresponding rows of $(x_i^{m_j})$ are $(x^{m_j})_j$ and $((-1)^{m_j} x^{m_j})_j = (-1)^{\delta}(x^{m_j})_j$: proportional, so the determinant vanishes. (No appeal to Lemma 4 is needed.) □

Convention 6 (orbits)**.** Fix $(v,u) \in \Sigma_{\lambda,f}$ and $r \ge 1$. By Corollary 5 only the *transversal* r-subsets contribute to (1), and these are indexed by triples (K, J, ε) with $K \subseteq [\lambda]$, $J \subseteq [f]$, $|K| + |J| = r$ and $\varepsilon \in \{\pm 1\}^K$:

$$T(K, J, \varepsilon) = \{a_i := \varepsilon_i v_i \;:\; i \in K\} \cup \{u_j : j \in J\}.$$

For fixed (K, J) the $2^{|K|}$ subsets $T(K, J, \cdot)$ form one orbit of the pair-flipping group $(\mathbb{Z}/2)^K \subseteq \mathrm{Sym}(M)$. Put

$$L := |K|, \qquad s := f - |J|, \qquad N := L + s.$$

Since $|K| + |J| = r$ we have, for *every* orbit,

$$L - s \;=\; |K| - f + |J| \;=\; r - f. \tag{3}$$

Equation (3) is the whole reason the threshold is $r \ge f + 2$: it says $r \ge f + 2$ holds if and only if $L \ge s + 2$ for every orbit simultaneously.

4 The orbit factorisation

Write $R(a,b) := (ab - 1)(a + b)$, and for $\varepsilon \in \{\pm 1\}^K$ write $\eta(\varepsilon) := \prod_{i \in K} \varepsilon_i$.

Lemma 7 (orbit factorisation)**.** *Let $(v,u) \in \Sigma^{\circ}_{\lambda,f}$, let all m_j be of parity δ, and fix (K, J) as in Convention 6. Then there is a constant $C_{K,J} \in k^{\times}$, independent of ε, with*

$$B\big(T(K,J,\varepsilon)\big)\, A\big(T(K,J,\varepsilon)\big) \;=\; C_{K,J} \cdot \eta(\varepsilon)^{\delta} \cdot \prod_{\substack{i<i' \\ i,i' \in K}} \frac{1}{R(a_i, a_{i'})} \cdot \prod_{i \in K} \prod_{p \in [f] \setminus J} \frac{1}{R(a_i, -u_p)}. \tag{4}$$

Proof. Write $T = T(K, J, \varepsilon)$ and $S = M \setminus T$, so that

$$S = \{-a_i : i \in K\} \;\sqcup\; \{v_k, -v_k : k \in [\lambda] \setminus K\} \;\sqcup\; \{u_p : p \in [f] \setminus J\}.$$

Throughout, $\doteq$ means "equal up to a nonzero factor not depending on ε". Flipping ε_i replaces a_i by $-a_i$ and fixes a_i^2; so any expression in the a_i^2, the v_k and the u_p is ε-free. I collect the factors of $N(T)A(T)/D(T)$ by the pair of elements they involve.

Singletons. $\prod_{x\in T}(x^2-1)$ from $N(T)$ is ε-free. $s_\mu(T^2)$ from Lemma 4 is ε-free, because $T^2 = \{v_i^2\}_{i\in K} \cup \{u_j^2\}_{j\in J}$. The factor $(\prod_{x\in T} x)^\delta$ of Lemma 4 contributes $\eta(\varepsilon)^\delta$, and the factor $\prod_{x\in T} x$ in $D(T)$ contributes $\eta(\varepsilon)^{-1}$; the sub-product of $\prod_{x\in T}\prod_{t\in S}(x-t)$ with $x = a_i,\ t = -a_i$ is $\prod_{i\in K} 2a_i \doteq \eta(\varepsilon)$, contributing a second $\eta(\varepsilon)^{-1}$. As $\eta = \pm 1$, these three give $\eta^{\delta-2} = \eta^\delta$. The sub-product with $x = a_i,\ t = -a_i$ of $\prod_{x\in T}\prod_{t\in S}(xt-1)$ is $\prod_i(-a_i^2-1)$, ε-free.

Pairs $i < i'$ in K. Put $P = a_i a_{i'}$, so P^2 is ε-free. The contributing factors are: $(a_i + a_{i'})$ from Lemma 4; $(P-1)$ from $\prod_{\{x,y\}\subseteq T}(xy-1)$; $(-P-1)^2$ from $\prod_{x\in T}\prod_{t\in S}(xt-1)$ at $(x,t) = (a_i, -a_{i'})$ and $(a_{i'}, -a_i)$; and $(a_i + a_{i'})^2$ in $D(T)$ from $\prod_{x\in T}\prod_{t\in S}(x-t)$ at the same two ordered pairs. The net factor is

$$\frac{(a_i+a_{i'})\,(P-1)\,(P+1)^2}{(a_i+a_{i'})^2} \;=\; \frac{(P^2-1)(P+1)}{a_i+a_{i'}} \;\doteq\; \frac{P+1}{a_i+a_{i'}} \;=\; \frac{P^2-1}{(P-1)(a_i+a_{i'})} \;\doteq\; \frac{1}{R(a_i,a_{i'})}.$$

Pairs $i \in K$, $j \in J$. The factors are $(a_i + u_j)$ (Lemma 4), $(a_i u_j - 1)$ (from $\prod_{\{x,y\}\subseteq T}$), $(-a_i u_j - 1)$ (from $\prod_{x\in T}\prod_{t\in S}(xt-1)$ at $x = u_j, t = -a_i$), and $(u_j + a_i)$ in $D(T)$ (from $\prod_{x\in T}\prod_{t\in S}(x-t)$ at $x = u_j, t = -a_i$). The net factor is $-(a_i^2u_j^2 - 1)$, which is ε-free. (These pairs drop out entirely; this is why J enters only through its complement.)

Pairs $i \in K$, $p \notin J$. The factors are $(a_i u_p - 1)$ (from $\prod_{x\in T}\prod_{t\in S}(xt-1)$) and $(a_i - u_p)$ in $D(T)$. The net factor is

$$\frac{a_iu_p-1}{a_i-u_p} = \frac{a_i^2u_p^2-1}{(a_iu_p+1)(a_i-u_p)} \doteq \frac{1}{(a_iu_p+1)(a_i-u_p)} = \frac{-1}{\big(a_i(-u_p)-1\big)\big(a_i+(-u_p)\big)} \doteq \frac{1}{R(a_i,-u_p)}.$$

Pairs $i \in K$, $k \notin K$. Both v_k and $-v_k$ lie in S, so these factors occur in $\pm$ pairs: $(a_iv_k-1)(-a_iv_k-1) = -(a_i^2v_k^2-1)$ and $(a_i - v_k)(a_i + v_k) = a_i^2 - v_k^2$, both ε-free. *Untouched pairs contribute nothing*; this is why λ does not appear in the final criterion.

Everything else (e^r; pairs inside J; pairs $j \in J$ with $k \notin K$ or $p \notin J$) involves no a_i and is ε-free.

Multiplying the four net contributions gives (4). Each factor of $C_{K,J}$ is a difference or product of distinct elements of M, or of the form $w_aw_b - 1$, and so is nonzero because M is admissible and reciprocal-free; hence $C_{K,J} \neq 0$. □

Verified exactly, 63/63 orbits over 10 cells with $\lambda \le 4$, $f \le 3$ (`t4_bridge.py`): the quotient of $B(T)A(T)$ by the displayed product takes a single value on each orbit.

Lemma 8 (linearisation)**.** *Put $\sigma(x) = x - x^{-1}$. Then $R(a,b) = ab\,\big(\sigma(a) + \sigma(b)\big)$, and $\sigma(-x) = -\sigma(x)$.*

Proof. $ab(\sigma(a)+\sigma(b)) = ab\big(a - a^{-1} + b - b^{-1}\big) = a^2b + ab^2 - b - a = (ab-1)(a+b)$. □

Corollary 9 (the orbit sum as a sign-twisted Cauchy sum)**.** *In the situation of Lemma 7, set*

$$x_i = \sigma(v_i)\ (i \in K), \qquad c_p = \sigma(-u_p) = -\sigma(u_p)\ (p \in [f] \setminus J), \qquad \nu = N - 1 + \delta.$$

Then, for a nonzero constant $C'_{K,J}$ independent of ε,

$$\sum_{\varepsilon\in\{\pm1\}^K} B(T(K,J,\varepsilon))A(T(K,J,\varepsilon)) \;=\; C'_{K,J}\sum_{\varepsilon\in\{\pm1\}^L}\eta(\varepsilon)^\nu \prod_{\alpha<\beta\le L}\frac{1}{\varepsilon_\alpha x_\alpha+\varepsilon_\beta x_\beta}\prod_{\alpha\le L}\prod_{p\le s}\frac{1}{\varepsilon_\alpha x_\alpha + c_p}. \quad (5)$$

Proof. Form the graph G on the N vertices $K \sqcup ([f]\setminus J)$, with an edge for every pair having at least one endpoint in K; assign the value a_i to $i\in K$ and $-u_p$ to $p\notin J$. The product in (4) is $\prod_{\text{edges}} R^{-1}$. By Lemma 8, $\prod_{\text{edges}} R = \prod_{\text{vertices}}(\text{value})^{\deg}\cdot\prod_{\text{edges}}(\sigma_\alpha+\sigma_\beta)$, and $\deg = N-1$ at a vertex of K and $\deg = L$ at a vertex of $[f]\setminus J$. Since $\sigma(\varepsilon_i v_i)=\varepsilon_i\sigma(v_i)=\varepsilon_i x_i$ and $\prod_{i\in K} a_i^{N-1} = \eta^{N-1}\prod_i v_i^{N-1}$, the right-hand side of (4) equals a constant times $\eta^\delta\eta^{-(N-1)}\prod_{\text{edges}}(\varepsilon_\alpha x_\alpha+\varepsilon_\beta x_\beta)^{-1}$, and $\eta^{\delta-(N-1)}=\eta^\nu$ as $\eta=\pm1$. Summing over ε gives (5). The denominators are nonzero: $\varepsilon_\alpha x_\alpha+\varepsilon_\beta x_\beta = R(a_\alpha,a_\beta)/(a_\alpha a_\beta)$ is nonzero because $a_\alpha a_\beta\neq 1$ (reciprocal-freeness) and $a_\alpha+a_\beta\neq 0$ (α,β lie in different negation pairs); and $\varepsilon_\alpha x_\alpha + c_p = R(a_\alpha,-u_p)/(-a_\alpha u_p)$ is nonzero because $a_\alpha u_p\neq -1$ (apply reciprocal-freeness to $-a_\alpha\in M$ and u_p) and $a_\alpha\neq u_p$ (distinctness). □

5 The core identity

Proposition 10 (core identity)**.** *Let $L,s\ge 0$, $N=L+s$, and $\nu\in\mathbb{Z}$. Let $x_1,\dots,x_L,c_1,\dots,c_s$ be independent indeterminates over $\mathbb{Q}$ and set*

$$Z \;=\; \sum_{\varepsilon\in\{\pm1\}^L}\Big(\prod_{\alpha=1}^{L}\varepsilon_\alpha\Big)^\nu \prod_{1\le\alpha<\beta\le L}\frac{1}{\varepsilon_\alpha x_\alpha+\varepsilon_\beta x_\beta}\prod_{\alpha=1}^{L}\prod_{p=1}^{s}\frac{1}{\varepsilon_\alpha x_\alpha + c_p}.$$

Then $Z=0$ in each of the two cases

$$\textit{(i) } L\ge s+2; \qquad\qquad \textit{(ii) } L\ge s+1 \textit{ and } \nu \textit{ odd.}$$

Lemma 11 (parity of Z in each free variable)**.** *With the notation of Proposition 10, $Z|_{x_\alpha\mapsto -x_\alpha} = (-1)^\nu Z$ for every $\alpha\le L$.*

Proof. Take $\alpha=1$. Substituting $-x_1$ for x_1 in the summand indexed by ε replaces every occurrence of $\varepsilon_1 x_1$ by $(-\varepsilon_1)x_1$ and touches nothing else, so that summand becomes $\eta(\varepsilon)^\nu$ times the ε-independent-of-x_1 part of the summand indexed by ε', where $\varepsilon'_1=-\varepsilon_1$ and $\varepsilon'_\beta=\varepsilon_\beta$ for $\beta>1$. Since $\eta(\varepsilon)=-\eta(\varepsilon')$, that summand equals $(-1)^\nu$ times the ε'-summand. As $\varepsilon\mapsto\varepsilon'$ is a bijection of $\{\pm1\}^L$, summing gives $Z(-x_1)=(-1)^\nu Z(x_1)$. □

Proof. Put

$$D \;=\; \prod_{1\le\alpha<\beta\le L}(x_\alpha^2-x_\beta^2)\cdot\prod_{\alpha=1}^{L}\prod_{p=1}^{s}(c_p^2-x_\alpha^2), \qquad \Delta \;=\; \prod_{1\le\alpha<\beta\le L}(x_\alpha^2-x_\beta^2).$$

Each summand's denominator is a product of distinct irreducible linear forms, each dividing D exactly once: $\varepsilon_\alpha x_\alpha+\varepsilon_\beta x_\beta$ is $\pm(x_\alpha\pm x_\beta)$, and $\varepsilon_\alpha x_\alpha+c_p$ is $c_p\pm x_\alpha$. Hence $P:=ZD$ is a polynomial, and it suffices to show $P=0$.

Step 1 (degree in x_1). Exactly $N-1$ factors of each summand involve x_1, namely the $L-1$ factors $\varepsilon_1x_1+\varepsilon_\beta x_\beta$ ($2\le\beta\le L$) and the s factors $\varepsilon_1x_1+c_p$; each is of degree 1 in x_1, and no

numerator involves x_1. So each summand is $O\big(x_1^{-(N-1)}\big)$ as $x_1 \to \infty$, and therefore so is Z. Since $\deg_{x_1} D = 2(L-1) + 2s$,

$$\deg_{x_1} P \;\le\; 2(L-1) + 2s - (N-1) \;=\; L + s - 1 \;=\; N - 1. \tag{6}$$

Step 2 (divisibility by Δ). Fix $\alpha < \beta \le L$ and $\varsigma \in \{\pm 1\}$, and let $\ell = x_\alpha - \varsigma x_\beta$, an irreducible linear form, with $V = \{\ell = 0\}$. For a summand indexed by ε, the factor $\varepsilon_\alpha x_\alpha + \varepsilon_\beta x_\beta$ vanishes on V precisely when $\varepsilon_\alpha \varsigma + \varepsilon_\beta = 0$, i.e. when $\varepsilon_\alpha \varepsilon_\beta = -\varsigma$; no other factor vanishes at a generic point of V. Let ι flip both ε_α and ε_β. Then ι preserves the set of singular indices (it preserves $\varepsilon_\alpha \varepsilon_\beta$) and is fixed-point free on it, so the singular summands are partitioned into ι-pairs. Write a singular summand as $T(\varepsilon) = g_\varepsilon / (\varepsilon_\alpha x_\alpha + \varepsilon_\beta x_\beta)$ with g_ε regular at the generic point of V. Put $A = \varepsilon_\alpha x_\alpha$ and $B = \varepsilon_\beta x_\beta$, so $B = -A$ on V. Comparing g_ε with $g_{\iota\varepsilon}$ factor by factor:

- $\prod \varepsilon$ is unchanged by ι (two flips), so the sign η^ν is unchanged;
- for $\gamma \notin \{\alpha, \beta\}$, $\gamma \le L$, the pair of factors $(A + X)(B + X)$ with $X = \varepsilon_\gamma x_\gamma$ becomes $(-A + X)(-B + X)$; on V both equal $X^2 - A^2$;
- likewise $(A + c_p)(B + c_p)$ becomes $(-A + c_p)(-B + c_p)$, both equal to $c_p^2 - A^2$ on V;
- factors involving neither α nor β are unchanged.

Hence $g_\varepsilon = g_{\iota\varepsilon}$ on V, and since $\varepsilon_\alpha x_\alpha + \varepsilon_\beta x_\beta = \pm \ell$,

$$T(\varepsilon) + T(\iota\varepsilon) = \frac{g_\varepsilon - g_{\iota\varepsilon}}{\pm \ell}$$

is regular at the generic point of V, because ℓ divides the numerator. Summing over the ι-pairs, Z is regular on V. As D vanishes to order exactly one on V, $P = ZD$ vanishes on V, i.e. $\ell \mid P$. Running over all $\alpha < \beta \le L$ and both ς gives $2\binom{L}{2}$ pairwise non-associate irreducible linear forms dividing P, whence

$$\Delta \;=\; \prod_{\alpha < \beta \le L} (x_\alpha - x_\beta)(x_\alpha + x_\beta) \;\Big|\; P.$$

Step 3 (divisibility by $\prod_\alpha x_\alpha$ when ν is odd). D is even in each x_α, so by Lemma 11 $P = ZD$ satisfies $P|_{x_\alpha \mapsto -x_\alpha} = (-1)^\nu P$. If ν is odd then P is odd in each x_α, so $P|_{x_\alpha = 0} = 0$ and $x_\alpha \mid P$. No x_γ divides any $x_\alpha \pm x_\beta$, so $x_1, \dots, x_L$ together with the $2\binom{L}{2}$ forms of Step 2 are pairwise non-associate irreducibles, and

$$\Delta \cdot \textstyle\prod_{\alpha=1}^{L} x_\alpha \;\Big|\; P \qquad (\nu \text{ odd}).$$

Step 4 (the squeeze). $\deg_{x_1} \Delta = 2(L-1)$, since exactly the $L - 1$ factors $x_1^2 - x_\beta^2$ involve x_1. Suppose $P \neq 0$.

In case (i), $\Delta \mid P$ forces $\deg_{x_1} P \ge 2(L-1)$, so with (6)

$$2(L-1) \;\le\; N - 1 \;=\; L + s - 1, \qquad \text{i.e.} \qquad L \le s + 1,$$

contradicting $L \ge s + 2$.

In case (ii), ν is odd, so Step 3 gives $\deg_{x_1} P \ge 2(L-1) + 1$ and (6) gives

$$2(L-1) + 1 \;\le\; L + s - 1, \qquad \text{i.e.} \qquad L \le s,$$

contradicting $L \ge s + 1$. In both cases $P = 0$, hence $Z = 0$. □

Remark 12. Both squeezes are exactly sharp: $2(L-1) > L+s-1 \iff L \geq s+2$ and $2(L-1)+1 > L+s-1 \iff L \geq s+1$, so no slack is discarded and no case is left over. Note that Step 2 is *not* a sign-reversing involution on the sum — ι reverses signs only on the hyperplane V, and only on the singular summands. The involution organises a residue cancellation, not the sum.

I found case (ii) by running `t10_proofinputs.py` as a control on case (i) and reading its four disagreements rather than its 40 agreements: every one was $L = s+1$ with ν odd. *The control that fires is the one carrying the information.*

Remark 13. Proposition 10 says nothing outside (i) and (ii), and indeed $Z \neq 0$ there: of the 44 cells with $L \leq 5$, $s \leq 4$, $N \leq 7$ and $\nu = N-1+\delta$, $\delta \in \{0,1\}$, exactly the 20 satisfying (i) or (ii) have $Z = 0$, measured at 12 random rational points each (`t10_proofinputs.py`).

6 The theorem

Theorem 14 (transversal cancellation)**.** *Let M be admissible with exactly λ negation pairs and $f = n - 2\lambda$ unpaired elements, and let $\theta \in \mathbb{Z}^r$ have ρ-shifts $m_j = \theta_j + j - 1$ pairwise distinct and* all of the same parity $\delta \in \{0,1\}$. *If*

$$r \;\geq\; f + 2 - \delta$$

then $\Psi^C_{(1^r),\theta}(M) = 0$. (So $r \geq f+2$ when all m_j are even, and the one step better $r \geq f+1$ when all m_j are odd.)

Proof. First assume M reciprocal-free, i.e. $(v,u) \in \Sigma^\circ_{\lambda,f}$. By Corollary 5 the non-transversal r-subsets contribute 0 to (1), so

$$\Psi^C_{(1^r),\theta}(M) = \sum_{(K,J)} \sum_{\varepsilon \in \{\pm 1\}^K} B\big(T(K,J,\varepsilon)\big) A\big(T(K,J,\varepsilon)\big),$$

the outer sum over $K \subseteq [\lambda]$, $J \subseteq [f]$ with $|K|+|J| = r$. Fix one (K,J) and let $L = |K|$, $s = f - |J|$, $N = L + s$. By Corollary 9 the inner sum is $C'_{K,J}$ times the specialisation of the rational function Z of Proposition 10 at $x_\alpha = \sigma(v_\alpha)$, $c_p = -\sigma(u_p)$, $\nu = N - 1 + \delta$; that specialisation is legitimate because no denominator vanishes there (end of the proof of Corollary 9). It remains to check that Proposition 10 applies. By (3), $L - s = r - f$ for this and every orbit.

- If $r \geq f+2$ then $L \geq s+2$: case (i).
- If $r = f+1$ (which is the remaining possibility only when $\delta = 1$) then $L = s+1$, so $N = 2s+1$ and $\nu = N - 1 + \delta = 2s + \delta$ is odd: case (ii).

Either way $Z = 0$ identically, so the inner sum is 0. Every orbit contributes 0, so $\Psi^C = 0$.

Now let M be merely admissible. By (1) and (2), $\Psi^C_{(1^r),\theta}$ is a rational function of (v,u) whose denominators are products of e^r, of differences $x - t$ of distinct elements of M, and of $\prod_{i<k}(x_k - x_i)$; all are nonzero on $\Sigma_{\lambda,f}$, so Ψ^C is regular there. $\Sigma_{\lambda,f}$ is a nonempty open subset of $\mathbb{A}^{\lambda+f}$, hence irreducible, and $\Sigma^\circ_{\lambda,f}$ is a nonempty open subset of it, hence dense. A regular function vanishing on a dense subset of an irreducible variety vanishes identically. So $\Psi^C = 0$ on all of $\Sigma_{\lambda,f}$. □

Corollary 15. $\Psi^C_{(1^r),\theta}(M) = 0$ *for every $\theta = (c, c+1, \ldots, c+r-1)$ with $c \in \mathbb{Z}$ (then $m_j = 2(j-1)+c$, of parity $\delta \equiv c$) as soon as $r \geq f + 2 - (c \bmod 2)$; in particular for $\theta = (0,1,\ldots,r-1)$ and $r \geq f+2$, which is the "if" half of Conjecture 1.*

Remark 16 (the sharpened threshold is new, and the old sweep could not have seen it). The 26/26 sweep behind Conjecture 1 used only $\theta = (0, 1, \dots, r-1)$, i.e. $m_j = 2(j-1)$, which is $\delta = 0$ throughout. It therefore reported $r = f+1$ as the clean boundary ("$r = f+1$ gave 0 zeros in every cell"), and that is correct — *for even* m. For odd m the boundary is one step lower. Measured: all m_j odd at $r = f+1$ gives $\Psi^C = 0$ in 160/160 draws ($\lambda \le 5$, $f \le 3$, both engines agreeing on every draw), while the two controls stay green — all m_j even at $r = f+1$ is nonzero 160/160, and all m_j odd at $r = f$ (one below the new threshold) is nonzero 120/120 (`t11_sharp.py`). The full criterion $r \ge f + 2 - \delta$ gives 0 in 368/368 swept draws.

Measured: 170/170 draws zero for same-parity m of arbitrary spacing, $\lambda \le 4$, $f \le 2$, $r \ge f+2$, $\delta \in \{0, 1\}$; and 126/126 for the shifted arithmetic progressions $\theta = (c, \dots, c+r-1)$, $c \in [-3, 3]$ (`t6_scope.py`, `t7_parity.py`). The grid of `t2_grid.py` agrees in 89/89 cells, $\lambda \le 4$, $f \le 3$, $r \le 6$.

Remark 17 (the parity hypothesis is load-bearing, and not necessary). A control run on m of *mixed* parity gives 149/170 nonzero (`t7_parity.py`), so the hypothesis is doing real work. It is not necessary, however: see §9.

7 The converse is false

Theorem 18. *The "only if" half of Conjecture 1 is false. Each of the following M is admissible and reciprocal-free, has $r < f+2$, has every term $B(T)A(T) \neq 0$, and has $\Psi^C_{(1^r),(0,\dots,r-1)}(M) = 0$:*

$$\begin{array}{lllll}
(a) & M = \{2, -2, 3, -3, \tfrac{1}{6}, -\tfrac{1}{6}\}, & \lambda = 3,\ f = 0,\ r = 1, & \prod_t w_t = -1; \\
(b) & M = \{2, -\tfrac{1}{2}\}, & \lambda = 0,\ f = 2,\ r = 1, & \prod_t w_t = -1; \\
(c) & M = \{3, -3, \tfrac{1}{9}\}, & \lambda = 1,\ f = 1,\ r = 1, & \prod_t w_t = -1; \\
(d) & M = \{2, -2, 5, -5, \tfrac{1}{100}\}, & \lambda = 2,\ f = 1,\ r = 1, & \prod_t w_t = +1.
\end{array}$$

Proof. Each is a finite exact computation, carried out in `fractions.Fraction` by two independent implementations of Ψ^C — the head-injection sum and the alternant expansion (1) — which agree on every one (`t1_refute.py`). Conceptually, all four are instances of Theorem C of the morning artifact (`thm:r1`) at $r = 1$, $\theta = (0)$, which gives

$$\Psi^C_{(1),(0)}(M) \;=\; 1 - \Big(\prod_{t \in M} t\Big)^{-2} \tag{7}$$

for every admissible M; so the zero locus at $r = 1$, $\theta = (0)$ is exactly $\{\prod_t w_t = \pm 1\}$, and each of (a)–(d) is a reciprocal-free point of the relevant stratum lying on it. □

Remark 19 (where the stratum meets $\prod_t w_t = \pm 1$). On $\Sigma^\circ_{\lambda,f}$ one has $\prod_t w_t = (-1)^\lambda \big(\prod_i v_i\big)^2 \prod_p u_p$, so $\{\prod_t w_t = \pm 1\}$ meets $\Sigma^\circ_{\lambda,f}$ in a nonempty hypersurface whenever $f \ge 1$ (solve for u_f), and for $f = 0$ whenever $\lambda \ge 3$ (example (a)). It does *not* meet it for $f = 0$ and $\lambda \le 2$: $\lambda = 1$ needs $v_1^2 = \pm 1$ and $\lambda = 2$ needs $v_1 v_2 = \pm 1$, and all of $v_1 = \pm 1$, $v_1 v_2 = 1$, $v_1(-v_2) = 1$ are excluded by reciprocal-freeness. So the smallest counterexamples are exactly where they are.

Remark 20 (one of the four was inside the swept domain). The sweep behind Conjecture 1 declared the domain $\lambda \le 4$, $f \le 3$, $n \le 7$, $r \in \{f+1, f+2, f+3\}$, 10 draws per cell. Witness (a) has $\lambda = 3$, $f = 0$, $n = 6$, $r = 1 = f+1$: that cell *was* visited, and reported "0 zeros". Nothing was wrong with the draws. $\{\prod_t w_t = \pm 1\}$ has codimension 1 in the stratum, and a random rational point misses it almost surely.

Corollary 21 (the corrected statement)**.** *Let θ have pairwise distinct ρ-shifts, all of parity δ. On $\Sigma_{\lambda,f}$:*

- *if* $r \geq f + 2 - \delta$ *then* $\Psi^C_{(1^r),\theta}$ *vanishes* identically *(Theorem 14);*
- *if* $r \leq f + 1 - \delta$ *then* $\Psi^C_{(1^r),\theta}$ *does not vanish identically, but its zero locus in* $\Sigma_{\lambda,f}$ *is still nonempty whenever* $\Sigma^\circ_{\lambda,f}$ *meets it — for* $r = 1$, $\theta = (0)$ *it is exactly* $\{\prod_t w_t = \pm 1\}$ *by (7), and Remark 19 says when that is nonempty.*

So the threshold of Conjecture 1 is right as a threshold for identical *vanishing — after the correction* $f + 2 \rightsquigarrow f + 2 - \delta$ *— and wrong as a pointwise criterion. The second bullet's non-vanishing is* computed*, not proved:* 0 *zeros in* 6 *independent draws in each of the* 41 *cells with* $r \leq f + 1$ *of* `t2_grid.py` *(*$\delta = 0$*), plus the two green controls of* `t11_sharp.py`*.*

8 A second, θ-independent mechanism

Theorem 14 needs the parity hypothesis. For $f = 0$ there is a cruder mechanism that needs nothing of θ at all, and it is worth recording because my first version of it was wrong and a control caught it.

Proposition 22. *Let* $f = 0$*, so* $M = -M$*. Then for every* θ*,*

$$\Psi^C_{(1^r),\theta}(M) \;=\; (-1)^{\sum_j m_j + \binom{r}{2}}\, \Psi^C_{(1^r),\theta}(M); \qquad \textit{hence } \Psi^C = 0 \textit{ if } \textstyle\sum_j m_j + \binom{r}{2} \textit{ is odd.}$$

Proof. $T \mapsto -T$ is a bijection of the r-subsets of M, and $M \setminus (-T) = -(M \setminus T)$.

N is invariant. Every factor of $N(T)$ in (2) is of the form $x^2 - 1$, $xy - 1$ or $xt - 1$ with both arguments ranging over sets that are negated together, so each factor is unchanged: $(-x)(-y) - 1 = xy - 1$.

D is invariant. $e = \prod_{t \in M} t$ is unchanged ($M = -M$). The factor $\prod_{x \in T} x$ acquires $(-1)^r$, and $\prod_{x \in T} \prod_{t \in M \setminus T} (x - t)$ acquires $(-1)^{r(n-r)}$, for a total of $(-1)^{r(n-r+1)}$. Since $f = 0$ forces $n = 2\lambda$ even, $r(n - r + 1) \equiv r(1 - r) = -r(r - 1) \equiv 0 \pmod 2$. So $D(-T) = D(T)$, and therefore $B(-T) = B(T)$.

A picks up the sign. Negating every x_i scales column j of $(x_i^{m_j})$ by $(-1)^{m_j}$, so $\det \mapsto (-1)^{\sum_j m_j} \det$, while $\prod_{i<k}(x_k - x_i) \mapsto (-1)^{\binom{r}{2}} \prod_{i<k}(x_k - x_i)$. Hence $A(-T) = (-1)^{\sum_j m_j + \binom{r}{2}} A(T)$.

Reindexing (1) by $T \mapsto -T$ gives the claim. □

Verified 140/140 (f=0, $\sum_j m_j + \binom{r}{2}$ odd, arbitrary m, $\lambda \leq 5$, $r \leq 5$); the control, $f = 0$ with $\sum_j m_j + \binom{r}{2}$ *even* and m of mixed parity, is nonzero 122/126 (`t9_corrected.py`).

Remark 23 (the control found the sign). I first stated Proposition 22 with the factor $(-1)^{\binom{r}{2}}$ alone, having silently assumed $\sum_j m_j$ even (true for $\theta = (0, \ldots, r-1)$, where $m_j = 2(j-1)$, and false in general). Run as a prediction for arbitrary θ it scored 26/60 — about half, which is the signature of an unmodelled sign. Inserting $(-1)^{\sum_j m_j}$ gives 140/140. *A sign specialised at the θ I had in mind read as a sign that was not there.*

9 What is still open

Remark 24 (the parity hypothesis is not necessary). Scanning all 16 parity patterns of m at $\lambda = 3$, $f = 1$, $r = 4$ (so $r \geq f + 2$), 6 independent draws per pattern, identical vanishing occurs for exactly six patterns (`t9_corrected.py`):

$$(0,0,0,0), \quad (1,1,1,1), \quad (0,1,1,1), \quad (1,0,1,1), \quad (1,1,0,1), \quad (1,1,1,0),$$

i.e. for $\#\{j : m_j \text{ even}\} \in \{0, 1, 4\}$ and not for $\{2, 3\}$. The first two are Theorem 14. The other four are a genuine further mechanism: all 6 draws give 0, so the vanishing is identical on the stratum, not accidental. The Laplace expansion of $\det(x_i^{m_j})$ along the even and odd columns writes $A(T)$ as a sum over $\binom{r}{\#\text{even}}$ terms, each carrying a *different* character of $(\mathbb{Z}/2)^K$, which is precisely why Lemma 7 does not apply — and presumably why one even column behaves like none. I have not pursued it.

Both #even $= 0$ and #even $= r$ are Theorem 14 (with $\delta = 1$ and $\delta = 0$), and at this cell ($r = 4 \geq f + 2 = 3$) the sharpened threshold buys nothing extra, so the four patterns with #even $= 1$ are genuinely outside everything proved here.

This is **Q578**: *for which parity profiles of* $(m_1, \ldots, m_r)$ *does the transversal cancellation hold, and with what threshold in* r*?* Measured answer at $\lambda = 3, f = 1, r = 4$: #even $\in \{0, 1, 4\}$. Proved: #even $\in \{0, r\}$, with threshold $r \geq f + 2$ for #even $= r$ and $r \geq f + 1$ for #even $= 0$.

Also still open, carried from this morning: Q575 (the zero locus on the reciprocal-free stratum for $r \geq 2$) and Q577 (the type-C contour formula). Today's Corollary 21 sharpens Q575: on the negation stratum with $r \leq f + 1$, the locus is cut out by (7) at $r = 1$ and is unknown for $r \geq 2$.

10 Verification

All arithmetic is exact: `fractions.Fraction` or SymPy over $\mathbb{Q}$; no floating point. Two independent implementations of Ψ^C are used throughout — the head-injection sum over ordered r-tuples (`code-1023-q559/psi.py`, eq. (1) of the morning artifact) and this session's `psi_pair` (alternant expansion with B in the pair form) — and they agree 350/350 over $n \leq 5$, $r \leq 4$, *with signs allowed and no hygiene exclusions* (`t0_crosscheck.py`). That last clause matters: the Q559 sampler excluded signs, so a cross-check inherited from it would have been blind on the axis this paper lives on.

file	what	result	domain
`t0_crosscheck.py`	engine vs. head sum	350/350	$n \leq 5$, $r \leq 4$, signed
`t1_refute.py`	Theorem 18	4 witnesses	exact, both engines
`t2_grid.py`	$r \geq f + 2 \Leftrightarrow$ identical 0	89/89 cells	$\lambda \leq 4, f \leq 3, r \leq 6$
`t3_classes.py`	every orbit sum vanishes alone	6 cells, all orbits	incl. $L = 4$, $L = 3, s = 1$
`t4_bridge.py`	Lemma 7, Lemma 8	63/63 orbits	$\lambda \leq 4$, $f \leq 3$
`t6_scope.py`	$\theta = (c, \ldots, c + r - 1)$	126/126	$c \in [-3, 3]$
`t7_parity.py`	Lemma 4; same-parity m	216/216; 170/170	$r \leq 4$; $\lambda \leq 4$
`t9_corrected.py`	Proposition 22; parity scan	140/140; 16 patterns	$\lambda \leq 5$
`t10_proofinputs.py`	Prop. 10 and its inputs	see below	$L \leq 5$, $s \leq 4$, $N \leq 7$
`t11_sharp.py`	$r \geq f + 2 - \delta$; 2 controls	160/160, 368/368; green	$\lambda \leq 5$, $f \leq 3$

`t10_proofinputs.py` is a positive control on the *proof*, not only on the statement. It forms the numerator $P = ZD$ in closed form using the exact cofactors $(x_\alpha^2 - x_\beta^2)/(\varepsilon_\alpha x_\alpha + \varepsilon_\beta x_\beta) = \varepsilon_\alpha(x_\alpha - \varepsilon_\alpha\varepsilon_\beta x_\beta)$ and $(c_p^2 - x_\alpha^2)/(\varepsilon_\alpha x_\alpha + c_p) = c_p - \varepsilon_\alpha x_\alpha$, and evaluates it exactly at random integer points. On the cells where $P \neq 0$ — so that the checks are not vacuous — it confirms Step 1's bound $\deg_{x_1} P \leq N - 1$ (obtained by finite differences in x_1) and Step 2's conclusion that P vanishes on every hyperplane $x_\alpha = \pm x_\beta$, with **0 violations of either** across all such cells. It also confirms that $P = 0$ exactly on the 20 cells satisfying (i) or (ii) and nowhere else, and that $2(L-1) > L+s-1 \iff L \geq s+2$ and $2(L-1)+1 > L+s-1 \iff L \geq s+1$ with no mismatch over $L \leq 6$, $s \leq 5$.

11 Grades

statement	grade	basis
Lemma 4, Cor. 5	`proved`	two-line proofs; 216/216
Lemma 7 (orbit factorisation)	`proved`	factor-by-factor; 63/63 orbits
Lemma 8 (linearisation)	`proved`	one line
Lemma 11 (parity of Z)	`proved`	reindexing $\varepsilon_1 \mapsto -\varepsilon_1$
Prop. 10 (core identity, both cases)	`proved`	degree–divisibility squeeze
Thm. 14 ($r \geq f + 2 - \delta$)	`proved`	Prop. 10 + density
Cor. 15	`proved`	specialisation of Thm. 14
Thm. 18 (converse false)	`proved`	4 exact witnesses, two engines
Prop. 22 ($f = 0$ sign)	`proved`	reindexing; 140/140
Cor. 21, second bullet	`computed`	41 cells + 280 control draws
Q578 parity profiles	`conjecture`	16 patterns at $r = 4$ only

No claim in this paper rests on any external source: every input is either proved here or is one of my own earlier results (`thm:alt` 2026-10-22, the pair form of B and `thm:r1` 2026-10-24 cycle 1), each re-verified by this session's independent engine.